PROJECT EVALUATION SHEET

Project Prototype Part 1 | Midterm

B.Sc. in Computer Science and Engineering (CSE) Program

Course code	CSC 3215	**Course title**	Web Technologies
Semester	Fall 2026-27	**Section**	J
Student Name			
Student ID		**Group No.**	

Course Outcome

CO3 [PO-c2]	**Develop a multi-tier web application for a targeted society**
AI Policy	***The use of AI tools is strongly discouraged. A student who still needs an AI tool must get the course teacher's consent in advance and must state which tool was used and the exact parts of the work it was used for. Every student must be able to explain all of the submitted work; any content generated by AI that the student cannot explain will be marked as Missing/Incorrect (0).***

Evaluation (Course Teacher's Use Only)

Assessment Attribute/Criteria	Component	Marks (0–5)	Criterion Mark
Front-End Design and Code Structure *(10 marks)*	Pages & Forms (HTML)		
	UI Design (CSS)		
	Code Structure (HTML/CSS/JS separation)		
Feature Implementation and Explanation *(10 marks)*	JS Form Validation		
	DOM Manipulation		
	VIVA (Code Explanation)		
		CO3 Total	

Project Details

Project Title	**MetroSheba — Online Metro Ticketing Platform for Dhaka MRT Line-6**

Project Description

Describe the targeted society or users, the problem the project solves and the main pages, forms and features of the prototype.

Targeted society and users. Daily commuters of the Dhaka Metro Rail (MRT Line-6, Uttara North to Motijheel) and the staff who administer ticket sales. Passengers currently queue at station counters and vending machines during peak hours, which wastes time and causes crowding. MetroSheba lets a commuter buy a ticket online from a phone or computer before reaching the station, and gives the authority a live view of sales.

Problem the project solves. There is no single place where a commuter can register, buy a metro ticket, see what the fare will be before paying, and keep a record of past journeys. Counter staff also have no quick way to see which routes sell most or how much revenue a day produced. The application solves both sides with one multi-tier system.

Architecture. The project follows a three-tier MVC structure. Views hold only HTML; styling lives in four external stylesheets; behaviour lives in ten external JavaScript files; controllers validate input and enforce the business rules; models hold every MySQL query using prepared statements. No page writes SQL and no model prints HTML.

Main pages and forms. Home (index.php); Registration form with OTP verification; Login form with a signed "Remember me" cookie; Passenger Dashboard with the ticket purchase form; My Tickets purchase history; Profile view with picture upload; Edit Profile form; Admin Dashboard for customer management; and the Admin Ticket Sales report.

JavaScript validation. Every form is validated in the browser before it is submitted — name (two words, no digits), email format, 11-digit mobile starting with 01, numeric NID, date of birth, password length and a 6-digit OTP. Each rule is repeated in PHP on the server, because the client-side copy can always be bypassed.

DOM manipulation. The station dropdowns and route map are generated at runtime; the fare and total update live as the stations or quantity change and the chosen leg of the route is highlighted; the ticket receipt, the purchase-history table, the customer table and the busiest-routes bar chart are all built with createElement and textContent rather than written into the page as HTML.

AJAX and MySQL. Ticket purchase, OTP delivery, ticket cancellation, profile-picture upload and all admin edit/delete actions run through XMLHttpRequest against PHP endpoints that return JSON, so no action reloads the page. Data is stored in a MySQL database (users and metroshebatickets) created by database/metrosheba.sql.

Fare rule. The fare rises by 10 BDT per station with a 20 BDT minimum, so Uttara North to Motijheel (14 stations) costs 140 BDT per ticket. The rule is defined once on the server and sent to the browser, and the server recalculates the total on purchase, so a tampered form cannot buy a cheaper ticket.

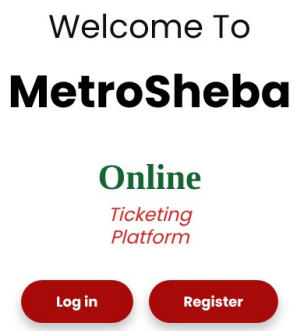
Snap of the UI

Screenshots of the main pages and forms of the prototype.

1. Home Page (index.php)

Landing page with Log in and Register actions.

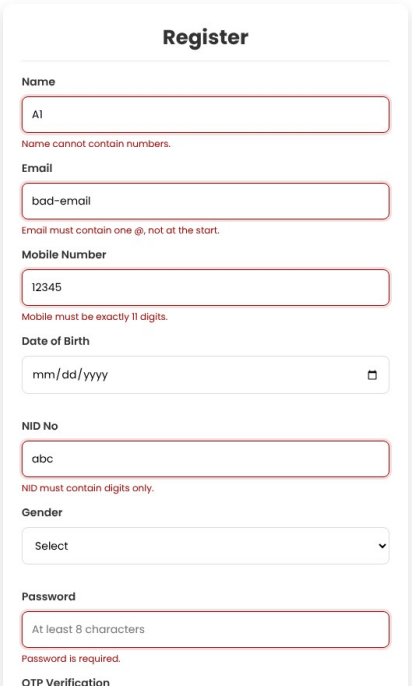




2. Registration Form — JavaScript Validation

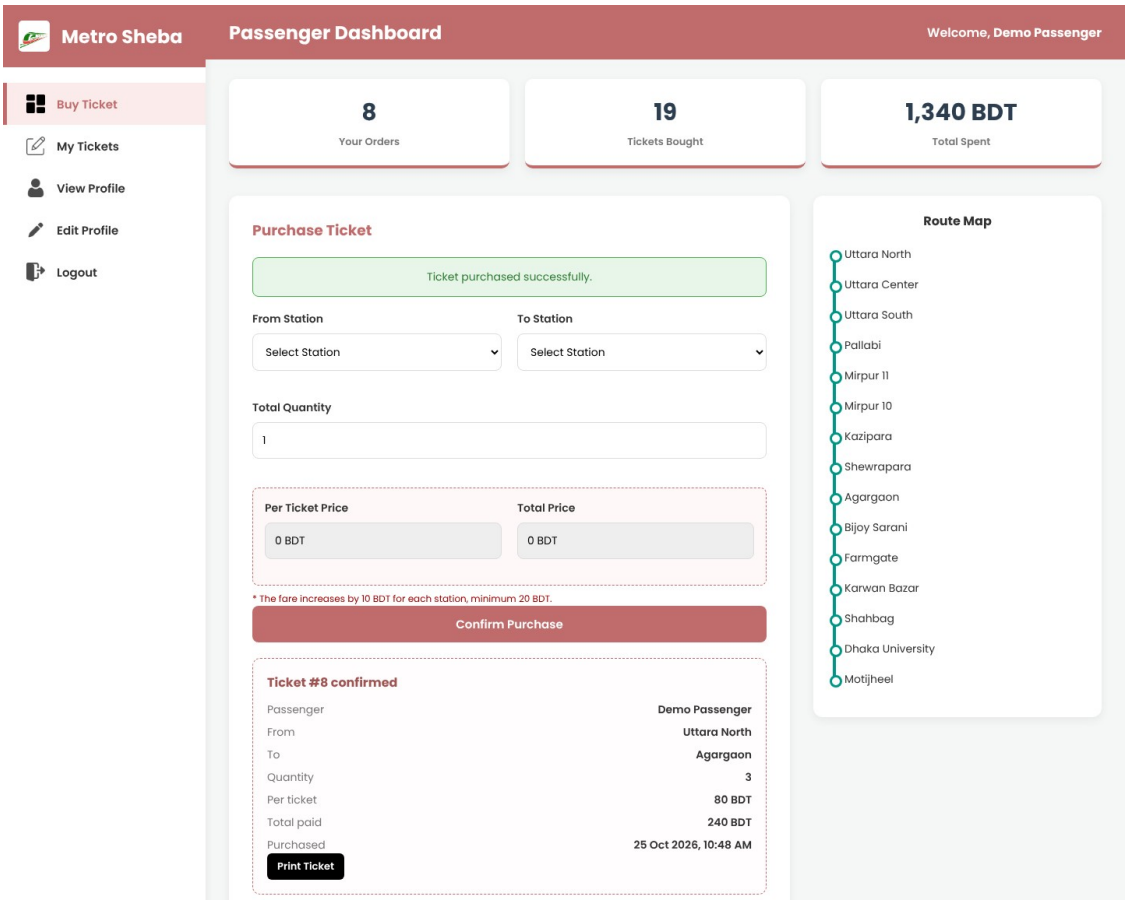
Each field is checked in the browser and the failing input is marked before the form can be submitted.





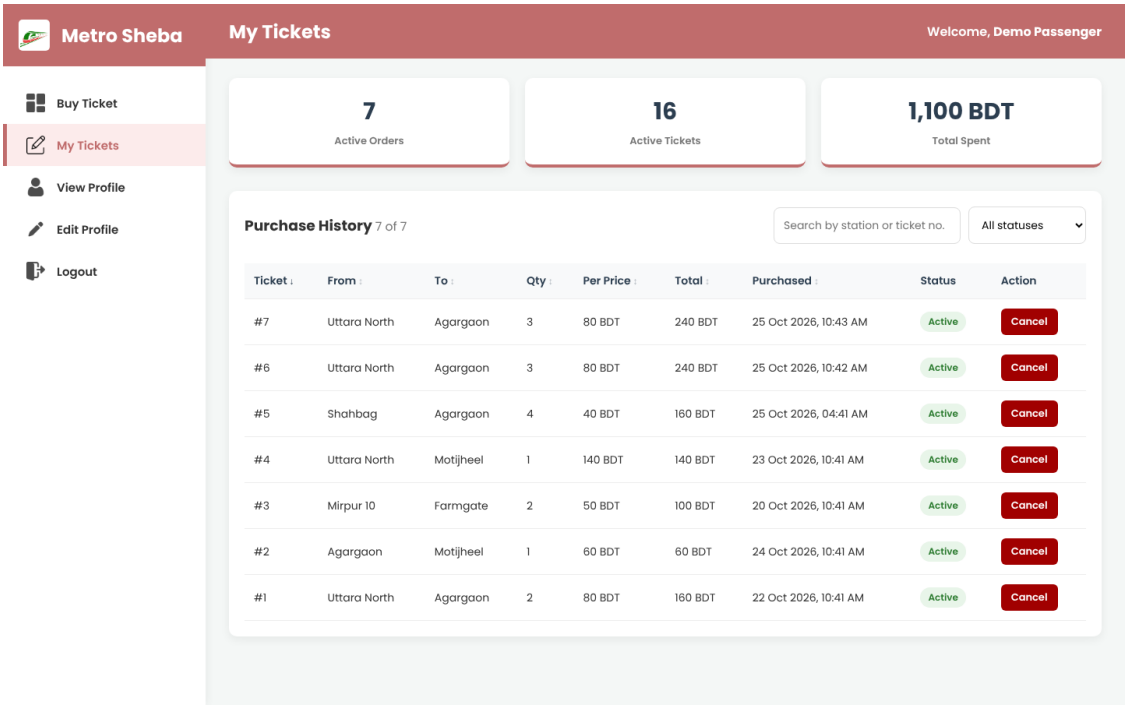
3. Passenger Dashboard — AJAX Purchase and DOM Receipt

The ticket is bought without reloading the page and the receipt is built in the DOM from the JSON reply.



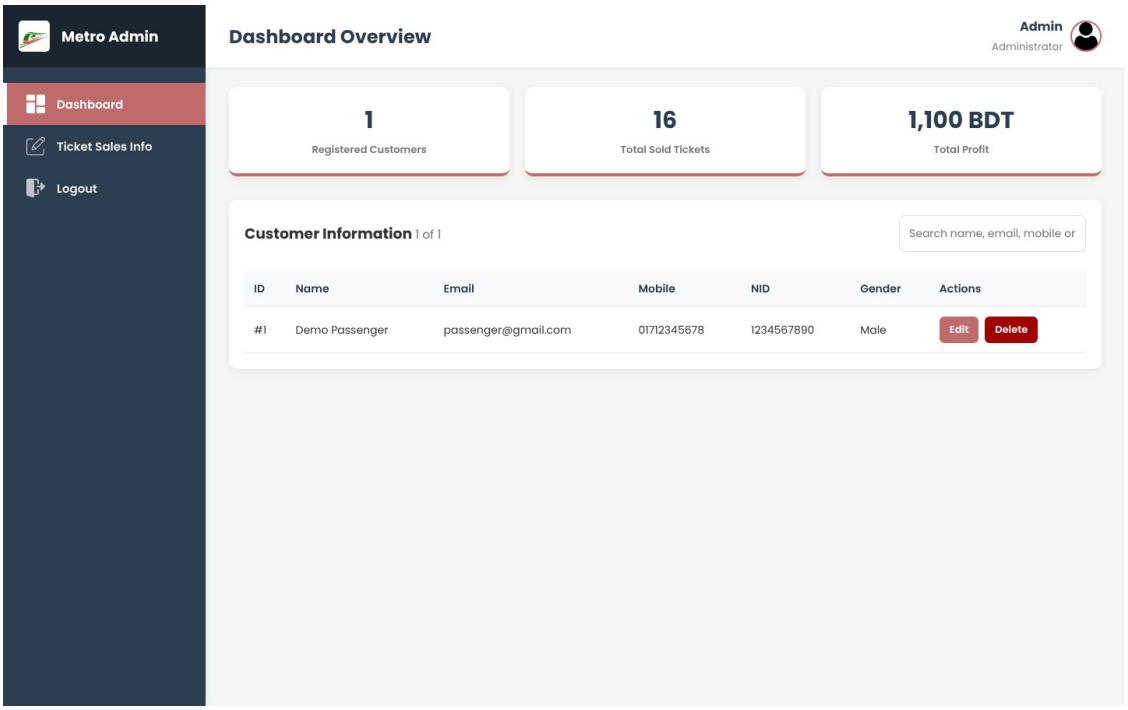
4. My Tickets — DOM Table with Search, Filter and Sorting

Purchase history loaded over AJAX; rows are created with createElement and can be searched, filtered, sorted and cancelled.



5. Admin Dashboard — Customer Management

Customers are listed over AJAX with live search; Edit and Delete act in place without leaving the page.



6. Admin Ticket Sales — Statistics, Chart and CSV Export

Sales statistics, a busiest-routes bar chart drawn with DOM nodes, sortable columns and a CSV export built in the browser.

